Math 550, S20 Name: Answer

Quiz 3

Instructions: **There are totally 3 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (3 points) Find all first and second-order partial derivatives of $f(x,y) = e^{x^2+y^2}$.

$$\frac{\partial f}{\partial x} = 2x e^{x^2+y^2} \qquad \frac{\partial f}{\partial y} = 2y e^{x^2+y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = 2e^{x^2+y^2} + 4x^2 e^{x^2+y^2} = e^{x^2+y^2}(2+4x^2)$$

$$\frac{\partial^2 f}{\partial y^2} = 2e^{x^2+y^2} + 4y^2 e^{x^2+y^2} = e^{x^2+y^2}(2+4y^2)$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = 4xy e^{x^2+y^2}$$

Problem 2. (3 points) Use the **Chain Rule** to find $D(f \circ \mathbf{g})$ when $f(x,y) = e^{x+2y}$ and $\mathbf{g}(s,t) = \left(\frac{s}{t}, \frac{t}{s}\right)$.

$$Df = [e^{x+2y} \quad 2e^{x+2y}] \qquad x = \frac{s}{t} \;\&\; y = \frac{t}{s}$$

$$Dg = \begin{bmatrix} \frac{1}{t} & -\frac{s}{t^2} \\ -\frac{t}{s^2} & \frac{1}{s} \end{bmatrix}$$

$$\Rightarrow D(f \circ g)(s,t) = Df\,Dg = [e^{x+2y} \quad 2e^{x+2y}] \begin{bmatrix} \frac{1}{t} & -\frac{s}{t^2} \\ -\frac{t}{s^2} & \frac{1}{s} \end{bmatrix}$$

$$= \left[e^{x+2y}\left(\frac{1}{t} - \frac{2t}{s^2}\right) \quad e^{x+2y}\left(-\frac{s}{t^2} + \frac{2}{s}\right)\right]$$

$$= \left[e^{\frac{s}{t}+\frac{2t}{s}}\left(\frac{1}{t} - \frac{2t}{s^2}\right) \quad e^{\frac{s}{t}+\frac{2t}{s}}\left(-\frac{s}{t^2} + \frac{2}{s}\right)\right]$$

Problem 3. (4 points) Given a function $f(x, y, z) = \sin(xy) + z^2$. Find the directional derivative of f at the point $P(1, 0, 1)$ along the direction of the vector $\mathbf{v} = (1, 2, 2)$. In what direction does f increase the most rapidly at P and what is the corresponding rate of change?

$$\nabla f(x,y,z) = (y\cos(xy), x\cos(xy), 2z)$$

$$\Rightarrow \nabla f(1,0,1) = (0, 1, 2)$$

$$u = \frac{v}{\|v\|} = \frac{(1,2,2)}{\sqrt{1^2+2^2+2^2}} = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

$$\Rightarrow D_u f\Big|_{(1,0,1)} = \nabla f(1,0,1) \cdot u = (0,1,2)\cdot\left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

$$= 0 + \frac{2}{3} + \frac{4}{3} = 2$$

f increase the most rapidly at $(1,0,1)$ in the direction of $\nabla f(1,0,1) = (0,1,2)$ and the corresponding rate of change is $\|\nabla f(1,0,1)\| = \sqrt{0^2+1^2+2^2} = \sqrt{5}$.